Name : **Class : B.E.** **Roll No :**

PRACTICAL NO. 05

Ensemble Learning: Voting, AdaBoost, Gradient Boosting and XGBoost on Iris

Aim

To train ensemble classifiers (hard Voting, AdaBoost, Gradient Boosting and XGBoost) on the Iris dataset and compare them using accuracy, macro precision, macro recall and macro F1-score.

Code and Output

In [1]:
```
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import (VotingClassifier, AdaBoostClassifier,
                              GradientBoostingClassifier,
                              RandomForestClassifier)
from sklearn.metrics import (accuracy_score, precision_score, recall_score,
                             f1_score, classification_report)
```

In [2]:
```
iris=load_iris()
X_train,X_test,y_train,y_test=train_test_split(
    iris.data,iris.target,test_size=.25,random_state=42,stratify=iris.target)
models={
 "Voting (hard)":VotingClassifier(estimators=[
     ("lr",make_pipeline(StandardScaler(),LogisticRegression(max_iter=2000))),
     ("dt",DecisionTreeClassifier(max_depth=3,random_state=42)),
     ("rf",RandomForestClassifier(n_estimators=150,random_state=42))],
     voting="hard"),
 "AdaBoost":AdaBoostClassifier(n_estimators=100,random_state=42),
 "Gradient Boosting":GradientBoostingClassifier(random_state=42)
}
try:
    from xgboost import XGBClassifier
    models["XGBoost"]=XGBClassifier(n_estimators=100,max_depth=3,
        learning_rate=.1,objective="multi:softprob",num_class=3,
        eval_metric="mlogloss",random_state=42)
except Exception as e:
    print("XGBoost unavailable; skipped:",e)
```

In [3]:
```
rows=[]
for name,model in models.items():
    model.fit(X_train,y_train); pred=model.predict(X_test)
    rows.append({"Model":name,"Accuracy":accuracy_score(y_test,pred),
```

```
        "Macro precision":precision_score(y_test,pred,average="macro",
                                          zero_division=0),
        "Macro recall":recall_score(y_test,pred,average="macro",
                                    zero_division=0),
        "Macro F1":f1_score(y_test,pred,average="macro",zero_division=0)})
    print("\n",name,"\n",classification_report(y_test,pred,
          target_names=iris.target_names,zero_division=0))
results=pd.DataFrame(rows).sort_values("Accuracy",ascending=False)
display(results.round(4))
```

```
 Voting (hard)
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        12
  versicolor       0.80      0.92      0.86        13
   virginica       0.91      0.77      0.83        13

    accuracy                           0.89        38
   macro avg       0.90      0.90      0.90        38
weighted avg       0.90      0.89      0.89        38


 AdaBoost
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        12
  versicolor       0.86      0.92      0.89        13
   virginica       0.92      0.85      0.88        13

    accuracy                           0.92        38
   macro avg       0.92      0.92      0.92        38
weighted avg       0.92      0.92      0.92        38


 Gradient Boosting
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        12
  versicolor       1.00      0.92      0.96        13
   virginica       0.93      1.00      0.96        13

    accuracy                           0.97        38
   macro avg       0.98      0.97      0.97        38
weighted avg       0.98      0.97      0.97        38


 XGBoost
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        12
  versicolor       0.86      0.92      0.89        13
   virginica       0.92      0.85      0.88        13

    accuracy                           0.92        38
   macro avg       0.92      0.92      0.92        38
weighted avg       0.92      0.92      0.92        38

              Model  Accuracy  Macro precision  Macro recall  Macro F1
2  Gradient Boosting    0.9737           0.9762        0.9744    0.9743
1           AdaBoost    0.9211           0.9246        0.9231    0.9230
3            XGBoost    0.9211           0.9246        0.9231    0.9230
0      Voting (hard)    0.8947           0.9030        0.8974    0.8968
```